Online Cake Ordering System

AIM

To design and develop an Online Cake Ordering System using HTML, CSS, and JavaScript that allows users to browse cakes, select categories and flavors, customize cakes, place orders, enter delivery details, make demo payments, track orders, and cancel orders. The system also provides an admin panel to manage cake products, inventory, and customer orders.

CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Cake Ordering System</title>

  <style>
    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #fff5f7;
      color: #333;
    }

    header {
      background: #b83268;
      color: white;
      text-align: center;
      padding: 22px;
    }

    nav {
```

```
  background: #78264b;
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
}

nav button {
  background: none;
  border: none;
  color: white;
  padding: 14px;
  cursor: pointer;
}

nav button:hover {
  background: #b83268;
}

.container {
  max-width: 1100px;
  margin: 25px auto;
  padding: 15px;
}

.page {
  display: none;
  background: white;
  padding: 22px;
  border-radius: 10px;
  box-shadow: 0 3px 10px #e5cbd4;
}

.page.active {
  display: block;
```

```
}

h2 {
  color: #b83268;
}

input, select, textarea {
  width: 100%;
  padding: 11px;
  margin: 7px 0 15px;
  border: 1px solid #ccc;
  border-radius: 5px;
}

.action {
  background: #b83268;
  color: white;
  border: none;
  padding: 11px 17px;
  border-radius: 5px;
  cursor: pointer;
  margin: 4px;
}

.action:hover {
  background: #78264b;
}

.cakes {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 18px;
  margin-top: 20px;
}
```

```
.cake-card {
  border: 1px solid #f0d2dd;
  border-radius: 9px;
  padding: 15px;
  text-align: center;
  background: #fffafb;
}

.cake-card img {
  width: 100%;
  height: 170px;
  object-fit: cover;
  border-radius: 7px;
}

.cake-card h3 {
  color: #b83268;
}

.message {
  color: #16834a;
  font-weight: bold;
}

.order-card {
  border: 1px solid #e7c8d4;
  padding: 15px;
  margin: 12px 0;
  border-radius: 7px;
}

footer {
  background: #78264b;
```

```
        color: white;
        text-align: center;
        padding: 15px;
        margin-top: 30px;
      }

      @media (max-width: 600px) {
        .container {
          margin: 10px;
          padding: 8px;
        }

        nav button {
          padding: 10px;
        }
      }
    </style>
</head>

<body>

<header>
    <h1>Sweet Treats - Online Cake Shop</h1>
    <p>Order Delicious Cakes for Every Occasion!</p>
</header>

<nav>
    <button onclick="showPage('home')">Home</button>
    <button onclick="showPage('register')">Register</button>
    <button onclick="showPage('login')">Login</button>
    <button onclick="showPage('catalog')">Browse Cakes</button>
    <button onclick="showPage('order')">Order Cake</button>
    <button onclick="showPage('payment')">Payment</button>
    <button onclick="showPage('track')">Track Order</button>
```

```
  <button onclick="showPage('history')">My Orders</button>
  <button onclick="showPage('admin')">Admin</button>
</nav>

<div class="container">

  <!-- Home Page -->
  <section id="home" class="page active">
    <h2>Welcome to Sweet Treats!</h2>
    <p>
      Find delicious cakes for birthdays, weddings, anniversaries,
      festivals, and other special occasions.
    </p>
    <p>Choose your favorite flavor, customize your cake, and place an order.</p>
    <button class="action" onclick="showPage('catalog')">
      Explore Cakes
    </button>
  </section>

  <!-- Registration Page -->
  <section id="register" class="page">
    <h2>Customer Registration</h2>

    <label>Full Name</label>
    <input type="text" id="regName" placeholder="Enter your name">

    <label>Email</label>
    <input type="email" id="regEmail" placeholder="Enter email">

    <label>Password</label>
    <input type="password" id="regPassword" placeholder="Create password">

    <button class="action" onclick="registerUser()">Register</button>
    <p id="registerMessage" class="message"></p>
```

```
</section>

<!-- Login Page -->
<section id="login" class="page">
  <h2>Customer Login</h2>

  <label>Email</label>
  <input type="email" id="loginEmail" placeholder="Enter email">

  <label>Password</label>
  <input type="password" id="loginPassword" placeholder="Enter password">

  <button class="action" onclick="loginUser()">Login</button>
  <p id="loginMessage" class="message"></p>
</section>

<!-- Cake Catalog -->
<section id="catalog" class="page">
  <h2>Our Cake Collection</h2>

  <label>Filter by Category</label>
  <select id="categoryFilter" onchange="displayCakes()">
    <option value="All">All Categories</option>
    <option value="Birthday">Birthday</option>
    <option value="Wedding">Wedding</option>
    <option value="Anniversary">Anniversary</option>
    <option value="Festival">Festival</option>
  </select>

  <label>Filter by Flavor</label>
  <select id="flavorFilter" onchange="displayCakes()">
    <option value="All">All Flavors</option>
    <option value="Chocolate">Chocolate</option>
    <option value="Vanilla">Vanilla</option>
```

```
      <option value="Strawberry">Strawberry</option>
      <option value="Black Forest">Black Forest</option>
    </select>

    <div id="cakeList" class="cakes"></div>
  </section>

  <!-- Order Cake -->
  <section id="order" class="page">
    <h2>Customize and Order Cake</h2>

    <label>Select Cake</label>
    <select id="cakeChoice"></select>

    <label>Size</label>
    <select id="cakeSize" onchange="updatePrice()">
      <option value="0.5">Half Kg</option>
      <option value="1">1 Kg</option>
      <option value="2">2 Kg</option>
      <option value="3">3 Kg</option>
    </select>

    <label>Flavor</label>
    <select id="customFlavor">
      <option>Chocolate</option>
      <option>Vanilla</option>
      <option>Strawberry</option>
      <option>Black Forest</option>
    </select>

    <label>Message on Cake</label>
    <input type="text" id="cakeMessage" placeholder="e.g. Happy Birthday">

    <label>Delivery Address</label>
```

```
  <textarea id="address" rows="3" placeholder="Enter full delivery address"></textarea>

  <label>Delivery Date</label>
  <input type="date" id="deliveryDate">

  <p>Estimated Price: ₹<strong id="price">500</strong></p>

  <button class="action" onclick="placeOrder()">Place Order</button>
  <p id="orderMessage" class="message"></p>
</section>

<!-- Payment Page -->
<section id="payment" class="page">
  <h2>Payment</h2>

  <label>Order ID</label>
  <input type="text" id="paymentOrderId" placeholder="Enter order ID">

  <label>Payment Method</label>
  <select id="paymentMethod">
    <option>UPI</option>
    <option>Debit Card</option>
    <option>Credit Card</option>
    <option>Net Banking</option>
    <option>Cash on Delivery</option>
  </select>

  <button class="action" onclick="makePayment()">Confirm Payment</button>
  <p id="paymentMessage" class="message"></p>
</section>

<!-- Track Order -->
<section id="track" class="page">
  <h2>Track Your Order</h2>
```

```
  <label>Order ID</label>
  <input type="text" id="trackOrderId" placeholder="Enter order ID">

  <button class="action" onclick="trackOrder()">Track Order</button>
  <div id="trackingResult"></div>
</section>

<!-- Order History -->
<section id="history" class="page">
  <h2>My Orders</h2>
  <button class="action" onclick="showOrders()">Refresh Orders</button>
  <div id="orderHistory"></div>
</section>

<!-- Admin Panel -->
<section id="admin" class="page">
  <h2>Admin Login</h2>

  <label>Username</label>
  <input type="text" id="adminUser" placeholder="Enter admin username">

  <label>Password</label>
  <input type="password" id="adminPassword" placeholder="Enter admin password">

  <button class="action" onclick="adminLogin()">Login as Admin</button>
  <p id="adminMessage" class="message"></p>

  <div id="adminPanel" style="display:none;">
    <h2>Add New Cake</h2>

    <label>Cake Name</label>
    <input type="text" id="newCakeName" placeholder="Enter cake name">
```

```
        <label>Category</label>
        <select id="newCakeCategory">
          <option>Birthday</option>
          <option>Wedding</option>
          <option>Anniversary</option>
          <option>Festival</option>
        </select>

        <label>Flavor</label>
        <select id="newCakeFlavor">
          <option>Chocolate</option>
          <option>Vanilla</option>
          <option>Strawberry</option>
          <option>Black Forest</option>
        </select>

        <label>Price per Kg (INR)</label>
        <input type="number" id="newCakePrice" min="1">

        <label>Image URL (optional)</label>
        <input type="url" id="newCakeImage" placeholder="Paste image URL">

        <button class="action" onclick="addCake()">Add Cake</button>

        <h2>Order Management</h2>
        <div id="adminOrders"></div>
      </div>
    </section>

</div>

<footer>
  <p>Sweet Treats | Online Cake Ordering System | Internet Programming Lab</p>
</footer>
```

```
<script>
  let users = [];
  let currentUser = null;
  let isAdmin = false;
  let orders = [];
  let nextOrderNumber = 1001;

  let cakes = [
    {
      id: 1,
      name: "Chocolate Cake",
      category: "Birthday",
      flavor: "Chocolate",
      price: 500,
      image: "https://images.unsplash.com/photo-1578985545062-
69928b1d9587?auto=format&fit=crop&w=600&q=80"
    },
    {
      id: 2,
      name: "Vanilla Cream Cake",
      category: "Wedding",
      flavor: "Vanilla",
      price: 450,
      image: "https://images.unsplash.com/photo-1565958011703-
44f9829ba187?auto=format&fit=crop&w=600&q=80"
    },
    {
      id: 3,
      name: "Strawberry Cake",
      category: "Anniversary",
      flavor: "Strawberry",
      price: 600,
      image: "https://images.unsplash.com/photo-1464349095431-
e9a21285b5f3?auto=format&fit=crop&w=600&q=80"
```

```
    },
    {
      id: 4,
      name: "Black Forest Cake",
      category: "Festival",
      flavor: "Black Forest",
      price: 550,
      image: "https://images.unsplash.com/photo-1606313564200-
e75d5e30476c?auto=format&fit=crop&w=600&q=80"
    }
  ];

  function showPage(id) {
    document.querySelectorAll(".page").forEach(page => {
      page.classList.remove("active");
    });

    document.getElementById(id).classList.add("active");

    if (id === "catalog") displayCakes();
    if (id === "order") updateCakeChoices();
    if (id === "history") showOrders();
    if (id === "admin" && isAdmin) renderAdminOrders();
  }

  function registerUser() {
    const name = document.getElementById("regName").value.trim();
    const email = document.getElementById("regEmail").value.trim();
    const password = document.getElementById("regPassword").value;

    if (!name || !email || !password) {
      alert("Please fill in all registration fields.");
      return;
    }
```

```
  if (users.some(user => user.email === email)) {
    alert("This email is already registered.");
    return;
  }

  users.push({ name, email, password });

  document.getElementById("registerMessage").textContent =
    "Registration successful. Please log in.";

  document.getElementById("regName").value = "";
  document.getElementById("regEmail").value = "";
  document.getElementById("regPassword").value = "";
}

function loginUser() {
  const email = document.getElementById("loginEmail").value.trim();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(item =>
    item.email === email && item.password === password
  );

  if (!user) {
    document.getElementById("loginMessage").textContent =
      "Invalid details. Register first or check your credentials.";
    return;
  }

  currentUser = user;

  document.getElementById("loginMessage").textContent =
    "Welcome, " + user.name + "! You are logged in.";
```

```
  showPage("catalog");
}

function displayCakes() {
  const category = document.getElementById("categoryFilter").value;
  const flavor = document.getElementById("flavorFilter").value;

  const filtered = cakes.filter(cake =>
    (category === "All" || cake.category === category) &&
    (flavor === "All" || cake.flavor === flavor)
  );

  const list = document.getElementById("cakeList");
  list.innerHTML = "";

  if (filtered.length === 0) {
    list.innerHTML = "<p>No cakes found for this selection.</p>";
    return;
  }

  filtered.forEach(cake => {
    list.innerHTML += `
      <div class="cake-card">
        <img src="${cake.image}" alt="${cake.name}"
          onerror="this.style.display='none'">
        <h3>${cake.name}</h3>
        <p>Category: ${cake.category}</p>
        <p>Flavor: ${cake.flavor}</p>
        <p>Price per Kg: ₹${cake.price}</p>
        <button class="action"
          onclick="chooseCake(${cake.id})">
          Order Now
        </button>
```

```
        </div>
      `;
    });
}

function updateCakeChoices() {
    const select = document.getElementById("cakeChoice");

    select.innerHTML = cakes.map(cake =>
      `<option value="${cake.id}">${cake.name} - ₹${cake.price}/Kg</option>`
    ).join("");

    updatePrice();
}

function chooseCake(id) {
    showPage("order");
    document.getElementById("cakeChoice").value = String(id);
    updatePrice();
}

function updatePrice() {
    const id = Number(document.getElementById("cakeChoice").value);
    const cake = cakes.find(item => item.id === id);

    if (!cake) return;

    const size = Number(document.getElementById("cakeSize").value);
    document.getElementById("price").textContent = cake.price * size;
}

document.getElementById("cakeChoice").addEventListener("change", updatePrice);

function placeOrder() {
```

```
if (!currentUser) {
  alert("Please register and log in before placing an order.");
  showPage("register");
  return;
}

const cakeId = Number(document.getElementById("cakeChoice").value);
const cake = cakes.find(item => item.id === cakeId);
const size = Number(document.getElementById("cakeSize").value);
const flavor = document.getElementById("customFlavor").value;
const message = document.getElementById("cakeMessage").value.trim();
const address = document.getElementById("address").value.trim();
const date = document.getElementById("deliveryDate").value;

if (!cake || !address || !date) {
  alert("Please select a cake and enter delivery address and date.");
  return;
}

if (date < new Date().toISOString().split("T")[0]) {
  alert("Please choose today or a future delivery date.");
  return;
}

const order = {
  id: "CAKE" + nextOrderNumber++,
  customer: currentUser.name,
  email: currentUser.email,
  cake: cake.name,
  size,
  flavor,
  message,
  address,
  date,
```

```
    total: cake.price * size,
    status: "Order Placed",
    paid: false
  };

  orders.push(order);

  document.getElementById("paymentOrderId").value = order.id;
  document.getElementById("trackOrderId").value = order.id;

  document.getElementById("orderMessage").textContent =
    "Order placed successfully! Your Order ID is " + order.id +
    ". Continue to payment.";

  showOrders();
}

function makePayment() {
  const id = document.getElementById("paymentOrderId").value.trim();
  const order = orders.find(item => item.id === id);

  if (!order) {
    document.getElementById("paymentMessage").textContent =
      "Order not found. Please enter a valid Order ID.";
    return;
  }

  if (order.status === "Cancelled") {
    document.getElementById("paymentMessage").textContent =
      "This order has been cancelled.";
    return;
  }

  if (order.paid) {
```

```
    document.getElementById("paymentMessage").textContent =
      "Demo payment has already been recorded.";
    return;
  }

  order.paid = true;
  order.status = "Payment Completed";

  document.getElementById("paymentMessage").textContent =
    "Demo payment recorded for ₹" + order.total +
    ". No real payment was made.";

  renderAdminOrders();
  showOrders();
}

function trackOrder() {
  const id = document.getElementById("trackOrderId").value.trim();
  const order = orders.find(item => item.id === id);
  const result = document.getElementById("trackingResult");

  if (!order) {
    result.innerHTML = "<p>Order not found. Check the Order ID.</p>";
    return;
  }

  result.innerHTML = `
    <div class="order-card">
      <h3>Order Tracking Details</h3>
      <p>Order ID: ${order.id}</p>
      <p>Cake: ${order.cake}</p>
      <p>Delivery Date: ${order.date}</p>
      <p>Status: <strong>${order.status}</strong></p>
      <p>Payment: ${order.paid ? "Demo payment completed" : "Pending"}</p>
```

```
      </div>
    `;
  }

  function showOrders() {
    const history = document.getElementById("orderHistory");

    const customerOrders = currentUser
      ? orders.filter(order => order.email === currentUser.email)
      : [];

    if (customerOrders.length === 0) {
      history.innerHTML = "<p>Please log in to view your orders.</p>";
      return;
    }

    history.innerHTML = customerOrders.map(order => `
      <div class="order-card">
        <h3>${order.cake}</h3>
        <p>Order ID: ${order.id}</p>
        <p>Size: ${order.size} Kg</p>
        <p>Total: ₹${order.total}</p>
        <p>Status: ${order.status}</p>
        <button class="action"
          onclick="document.getElementById('trackOrderId').value='${order.id}'; showPage('track');
trackOrder();">
          Track
        </button>
        ${order.status !== "Cancelled" ? `
          <button class="action" onclick="cancelOrder('${order.id}')">
            Cancel Order
          </button>` : ""}
      </div>
    `).join("");
```

```
}

function cancelOrder(id) {
  const order = orders.find(item => item.id === id);

  if (!order || order.status === "Cancelled") return;

  order.status = "Cancelled";

  alert("Demo order cancelled. No real refund has been processed.");

  showOrders();
  renderAdminOrders();
}

function adminLogin() {
  const username = document.getElementById("adminUser").value;
  const password = document.getElementById("adminPassword").value;

  if (username === "admin" && password === "admin123") {
    isAdmin = true;

    document.getElementById("adminMessage").textContent =
      "Admin login successful.";

    document.getElementById("adminPanel").style.display = "block";

    renderAdminOrders();
  } else {
    document.getElementById("adminMessage").textContent =
      "Invalid admin username or password.";
  }
}
```

```
function addCake() {
  if (!isAdmin) {
    alert("Please log in as admin first.");
    return;
  }

  const name = document.getElementById("newCakeName").value.trim();
  const category = document.getElementById("newCakeCategory").value;
  const flavor = document.getElementById("newCakeFlavor").value;
  const price = Number(document.getElementById("newCakePrice").value);
  const image = document.getElementById("newCakeImage").value.trim();

  if (!name || price <= 0) {
    alert("Enter a cake name and valid price.");
    return;
  }

  cakes.push({
    id: Date.now(),
    name,
    category,
    flavor,
    price,
    image: image || "https://placehold.co/600x400?text=Cake"
  });

  alert("Cake added successfully.");

  document.getElementById("newCakeName").value = "";
  document.getElementById("newCakePrice").value = "";
  document.getElementById("newCakeImage").value = "";

  displayCakes();
  updateCakeChoices();
```

```
}

function renderAdminOrders() {
  const panel = document.getElementById("adminOrders");

  if (!isAdmin) return;

  if (orders.length === 0) {
    panel.innerHTML = "<p>No orders available.</p>";
    return;
  }

  panel.innerHTML = orders.map(order => `
    <div class="order-card">
      <p><strong>Order:</strong> ${order.id}</p>
      <p><strong>Customer:</strong> ${order.customer}</p>
      <p><strong>Cake:</strong> ${order.cake}</p>
      <p><strong>Total:</strong> ₹${order.total}</p>
      <p><strong>Status:</strong> ${order.status}</p>
      <select id="status-${order.id}">
        <option>Order Placed</option>
        <option>Payment Completed</option>
        <option>Preparing</option>
        <option>Out for Delivery</option>
        <option>Delivered</option>
        <option>Cancelled</option>
      </select>
      <button class="action"
        onclick="updateOrderStatus('${order.id}')">
        Update Status
      </button>
    </div>
  `).join("");
}
```

```
  function updateOrderStatus(id) {
    if (!isAdmin) return;

    const order = orders.find(item => item.id === id);
    const status = document.getElementById("status-" + id).value;

    if (order) {
      order.status = status;
      alert("Order status updated.");
      renderAdminOrders();
      showOrders();
    }
  }

  displayCakes();
  updateCakeChoices();
</script>

</body>
</html>
```

